

Assignment 2

Question 1 (10 marks)

- (a) Give an algebraic proof that $k \cdot \binom{n}{k} = n \cdot \binom{n-1}{k-1}$.
- (b) Prove this result using a combinatorial argument.

Question 2 (10 marks)

Suppose that black stones and white stones are placed on a 4×4 grid in such a way that the following hold:

1. There is at least one black stone.
2. There are at least two white stones, at least one of which is on a corner cell.
3. Each grid cell has zero or one stone.
4. No two black stones are adjacent (diagonal *does not* count as adjacent).

For any configuration of stones, set $b_{ij} = 1$ if a black stone is on cell (i, j) and $w_{ij} = 1$ if a white stone is on cell (i, j) , where $i, j \in \{1, 2, 3, 4\}$. Write a formula f in **propositional logic** such that for any board configuration, f evaluates to true if and only if all four conditions above hold.

Since your final formula f will be quite long, please write several subformulas and **clearly explain** which condition(s) each subformula is describing. You may use the following notation.

$$\bigvee_{i=1}^n p_i = p_1 \vee p_2 \vee \dots \vee p_n \quad \text{and} \quad \bigwedge_{i=1}^n p_i = p_1 \wedge p_2 \wedge \dots \wedge p_n$$

Question 3 (10 marks)

Prove or disprove each statement, using $\mathbb{R}$ as the universe of discourse.

- (a) $\forall x. \forall y. \left(x < y \implies \frac{(x+y)}{2} < y \right)$
- (b) $\forall x. \forall y. \exists z. (zx > y).$
- (c) $\neg(\exists x. \forall y. (y \neq x \implies y < x)).$

Question 4 (10 marks)

Consider the 3D grid with integer coordinates. That is: a point (x, y, z) lies on the grid if and only if x, y and z are all integers.

Prove that if we take nine points on the grid then there exist two of the points whose average is also a point on the grid.

Hint: Use the Pigeonhole Principle.